

Calculus I (MA1110)

Comprehensive Rigorous Companion & Solutions Manual

FALL 2026 — INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD (IITH)

Complete Course Coverage:

- **Unit 1: The Real Number System ($\mathbb{R}$)** — Order Axioms, Completeness/Least Upper Bound (LUB) Property, Archimedean Property, Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$, Absolute Value & Triangle Inequalities.
 - **Unit 2: Sequences of Real Numbers** — Formal ε - N Convergence, Uniqueness of Limits, Limit Algebra, Monotone Convergence Theorem (MCT), Sandwich/Squeeze Theorem, Cauchy Criterion, Subsequences, and the Bolzano–Weierstrass Theorem.
 - **Unit 3: Infinite Series** — Convergence of Partial Sums, Divergence Test (n -th term), Telescoping Series, Geometric & p -Series, Comparison Tests, d’Alembert’s Ratio Test, Cauchy’s Root Test, Abel–Pringsheim Criterion, Alternating Series & Leibniz Test, Absolute vs. Conditional Convergence.
 - **Unit 4: Continuity of Functions** — Sequential and ε - δ Definitions and Equivalence Proof, Preservation of Signs, Intermediate Value Theorem (IVT) and Applications.
 - **Complete Solutions to all Exercises (1 through 19)** with complete analytical proofs.
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Synthesized from Lectures 1 through 9

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1 Introduction and Properties of the Real Numbers

1.1 What is Calculus and Why Do We Study It?

Calculus is the **mathematics of continuous change**. It provides the analytical and quantitative machinery to study phenomena where quantities evolve smoothly over time, space, or other continuous parameters.

Key Engineering and Scientific Applications:

- **Mechanical & Aerospace Engineering:** Instantaneous velocities, accelerations, rigid body dynamics, fluid dynamics, stress tensors, aerodynamic wing profile optimization.
- **Electrical & Computer Engineering:** Time-varying currents, electromagnetic field theory via Maxwell's equations, signal and image processing.
- **Computer Science & Machine Learning:** Loss landscape optimization, gradient descent, backpropagation algorithms in deep neural networks.
- **Civil & Chemical Engineering:** Structural deflection in load-bearing beams, reaction kinetics, mass transport, diffusion equations.

Calculus was developed independently in the late 17th century by **Sir Isaac Newton** (1642–1727) and **Gottfried Wilhelm Leibniz** (1646–1716).

1.2 The Order and Completeness Properties of $\mathbb{R}$

The real number system $\mathbb{R}$ contains the natural numbers $\mathbb{N}$, integers $\mathbb{Z}$, and rationals $\mathbb{Q}$:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c \subset \mathbb{C}.$$

A distinguishing feature of $\mathbb{R}$ is that $(\mathbb{R}, \leq)$ forms a **totally ordered field**.

- **Law of Trichotomy:** For any two real numbers $x, y \in \mathbb{R}$, exactly one of the following holds:

$$1. x = y, \quad 2. x < y, \quad 3. y < x.$$

Definition: Upper Bounds, Lower Bounds, Bounded Sets

Let $A \subset \mathbb{R}$ be a nonempty subset.

1. **Upper Bound:** A real number $\alpha \in \mathbb{R}$ is an *upper bound* of A if for each $x \in A$, $x \leq \alpha$. In quantifiers: $\forall x \in A (x \leq \alpha)$.
2. **Not Bounded Above:** A set A is not bounded above if: $\forall \alpha \in \mathbb{R}, \exists x \in A$ such that $x > \alpha$.
3. **Lower Bound:** A real number $\beta \in \mathbb{R}$ is a *lower bound* of A if for each $x \in A$, $\beta \leq x$. In quantifiers: $\forall x \in A (\beta \leq x)$.
4. **Bounded Set:** A is said to be *bounded* if it is both bounded above and bounded below, i.e., $\exists m, M \in \mathbb{R}$ such that $m \leq x \leq M$ for all $x \in A$.

Definition: Supremum (LUB) and Infimum (GLB)

Let $A \subset \mathbb{R}$ be a nonempty set.

1. $\alpha \in \mathbb{R}$ is the **Least Upper Bound (LUB)** or **Supremum** ($\alpha = \sup A$) if:
 - (i) α is an upper bound of A ($x \leq \alpha, \forall x \in A$).
 - (ii) If β is any upper bound of A , then $\alpha \leq \beta$. Equivalently: $\forall \varepsilon > 0, \exists x \in A$ such that $x > \alpha - \varepsilon$.
2. $\alpha \in \mathbb{R}$ is the **Greatest Lower Bound (GLB)** or **Infimum** ($\alpha = \inf A$) if:
 - (i) α is a lower bound of A ($\alpha \leq x, \forall x \in A$).
 - (ii) If β is any lower bound of A , then $\beta \leq \alpha$. Equivalently: $\forall \varepsilon > 0, \exists x \in A$ such that $x < \alpha + \varepsilon$.
3. If $\sup A \in A$, it is called the **Maximum** ($\max A$). If $\inf A \in A$, it is called the **Minimum** ($\min A$).

Theorem: Lower Bound is Less Than or Equal to Upper Bound (Theorem 1)

Let $A \subset \mathbb{R}$ be a nonempty set. If α is a lower bound of A and β is an upper bound of A , then $\alpha \leq \beta$.

Proof. Since A is nonempty, there exists at least one element $a \in A$. Since α is a lower bound of A , we have $\alpha \leq a$. Since β is an upper bound of A , we have $a \leq \beta$. By transitivity of the order relation in $\mathbb{R}$:

$$\alpha \leq a \leq \beta \implies \alpha \leq \beta.$$

□

Example: Supremum, Infimum, Maximum, and Minimum Examples

1. $A = (0, 1)$: $\sup A = 1, \inf A = 0$. Since $0, 1 \notin A$, $\max A$ and $\min A$ do not exist.
2. $A = [0, 1]$: $\sup A = 1 = \max A, \inf A = 0 = \min A$.
3. $A = (0, 1]$: $\sup A = 1 = \max A, \inf A = 0$ (no minimum since $0 \notin A$).
4. $A = \{1/n : n \in \mathbb{N}\} = \{1, 1/2, 1/3, \dots\}$: $\sup A = 1 = \max A, \inf A = 0$ (no minimum).
5. $A = \mathbb{N}$: $\inf A = 1 = \min A$; $\sup A$ does not exist in $\mathbb{R}$ (unbounded above).
6. $A = \{x \in \mathbb{R} : x^2 < 4\} = (-2, 2)$: $\sup A = 2, \inf A = -2$ (no max, no min).

Theorem: The Completeness Property of $\mathbb{R}$

Every nonempty subset $A \subset \mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$ ($\sup A \in \mathbb{R}$).

Note on $\mathbb{Q}$: The set of rationals $\mathbb{Q}$ does **not** satisfy the LUB property. For example, $A = \{x \in \mathbb{Q} : x^2 < 2\}$ is bounded above in $\mathbb{Q}$ (e.g., by 2), but $\sup A = \sqrt{2} \notin \mathbb{Q}$. This fundamental gap

motivates the construction of $\mathbb{R}$.

Theorem: Unboundedness of $\mathbb{N}$ in $\mathbb{R}$ (Theorem 2)

The set $\mathbb{N} = \{1, 2, 3, \dots\}$ is not bounded above in $\mathbb{R}$.

Proof. Suppose, for contradiction, that $\mathbb{N}$ is bounded above in $\mathbb{R}$. Since $\mathbb{N} \neq \emptyset$, by the Completeness Property of $\mathbb{R}$, $\mathbb{N}$ has a least upper bound, say $\alpha = \sup \mathbb{N} \in \mathbb{R}$.

Since $\alpha - 1 < \alpha$, $\alpha - 1$ cannot be an upper bound of $\mathbb{N}$. Therefore, there exists some $n \in \mathbb{N}$ such that:

$$n > \alpha - 1 \implies n + 1 > \alpha.$$

Since $n \in \mathbb{N}$, we also have $n + 1 \in \mathbb{N}$. Thus, $n + 1$ is an element of $\mathbb{N}$ strictly greater than the upper bound α , which is a contradiction.

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. $\square$

Theorem: Archimedean Property of $\mathbb{R}$ (Theorem 3)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ such that $nx > y$.

Proof. Suppose, for contradiction, that there exist $x > 0$ and $y \in \mathbb{R}$ such that $nx \leq y$ for all $n \in \mathbb{N}$.

Since $x > 0$, we can divide by x to obtain:

$$n \leq \frac{y}{x}, \quad \forall n \in \mathbb{N}.$$

This implies that the real number $\frac{y}{x}$ is an upper bound for the set of natural numbers $\mathbb{N}$. This directly contradicts Theorem 2 (unboundedness of $\mathbb{N}$).

Hence, there must exist $n \in \mathbb{N}$ such that $nx > y$. $\square$

Theorem: Density of Rational Numbers in $\mathbb{R}$ (Theorem 4)

Let $a, b \in \mathbb{R}$ with $a < b$. Then there exists a rational number $r \in \mathbb{Q}$ satisfying $a < r < b$.

Proof. Since $b > a$, we have $b - a > 0$. By the Archimedean Property (Theorem 3 with $x = b - a$ and $y = 1$), there exists $n \in \mathbb{N}$ such that:

$$n(b - a) > 1 \iff nb - na > 1.$$

Let $k = \lfloor na \rfloor$ (the largest integer $\leq na$). Define $m = k + 1 \in \mathbb{Z}$. By definition of floor:

$$k \leq na < k + 1 = m \implies na < m.$$

We claim that $m < nb$. Suppose, on the contrary, that $m \geq nb$. Then:

$$1 = (k + 1) - k = m - k \geq nb - na = n(b - a) > 1,$$

which gives $1 > 1$, an impossibility. Therefore, $m < nb$.

Combining the inequalities yields:

$$na < m < nb \implies a < \frac{m}{n} < b.$$

Since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, the number $r = \frac{m}{n}$ is rational. $\square$

Corollary: Density of Irrational Numbers in $\mathbb{R}$ (Corollary 1)

Let $a, b \in \mathbb{R}$ with $a < b$. Then there exists an irrational number $t \in \mathbb{R} \setminus \mathbb{Q}$ satisfying $a < t < b$.

Proof. Consider the real numbers $a - \sqrt{2}$ and $b - \sqrt{2}$. Since $a < b$, we have $a - \sqrt{2} < b - \sqrt{2}$. By the density of $\mathbb{Q}$ in $\mathbb{R}$ (Theorem 4), there exists a rational number $r \in \mathbb{Q}$ such that:

$$a - \sqrt{2} < r < b - \sqrt{2}.$$

Adding $\sqrt{2}$ across the inequality gives:

$$a < r + \sqrt{2} < b.$$

Let $t = r + \sqrt{2}$. We claim t is irrational. If $t \in \mathbb{Q}$, then $\sqrt{2} = t - r$ would be the difference of two rational numbers, and hence rational, which contradicts the known irrationality of $\sqrt{2}$. Thus $t \in \mathbb{R} \setminus \mathbb{Q}$, and $a < t < b$. $\square$

Definition: Properties of Absolute Value

For $x \in \mathbb{R}$, the absolute value is defined by $|x| = x$ if $x \geq 0$, and $-x$ if $x < 0$.

1. $|a| \geq 0$, and $|a| = 0 \iff a = 0$. Moreover, $-|a| \leq a \leq |a|$.
2. For $\varepsilon > 0$, $|a - b| < \varepsilon \iff a \in (b - \varepsilon, b + \varepsilon)$.
3. $|ab| = |a||b|$ and $|\frac{a}{b}| = \frac{|a|}{|b|}$ for $b \neq 0$.
4. **Triangle Inequality:** $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
5. **Reverse Triangle Inequality:** $||a| - |b|| \leq |a - b|$.
6. $\max\{a, b\} = \frac{a+b+|a-b|}{2}$ and $\min\{a, b\} = \frac{a+b-|a-b|}{2}$.

2 Sequences of Real Numbers

2.1 Definition of Convergence

A sequence of real numbers is a function $f : \mathbb{N} \rightarrow \mathbb{R}$, denoted $(x_n)_{n=1}^{\infty}$ or $(x_1, x_2, \dots)$.

Definition: Convergence of a Sequence (ε - N Criterion)

Let (x_n) be a sequence of real numbers and let $x \in \mathbb{R}$. We say that (x_n) converges to x (written $\lim_{n \rightarrow \infty} x_n = x$ or $x_n \rightarrow x$) if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, |x_n - x| < \varepsilon.$$

Geometric Meaning: For every $\varepsilon > 0$, the interval $(x - \varepsilon, x + \varepsilon)$ contains all terms x_n for $n \geq N$. That is, **all but finitely many terms** of the sequence lie inside $(x - \varepsilon, x + \varepsilon)$.

Definition: Divergence of a Sequence

A sequence (x_n) is **divergent** if it fails to converge to any real number $x \in \mathbb{R}$. In quantifiers:

$$\forall x \in \mathbb{R}, \exists \varepsilon_0 > 0 \text{ such that } \forall N \in \mathbb{N}, \exists n \geq N \text{ with } |x_n - x| \geq \varepsilon_0.$$

Theorem: Uniqueness of the Limit (Theorem 5)

If $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} x_n = y$, then $x = y$.

Proof. Let $\varepsilon > 0$ be arbitrary. Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that $|x_n - x| < \frac{\varepsilon}{2}$ for all $n \geq N_1$. Since $x_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that $|x_n - y| < \frac{\varepsilon}{2}$ for all $n \geq N_2$. Choose $N = \max\{N_1, N_2\}$. Then for any $n \geq N$, by the triangle inequality:

$$|x - y| = |(x - x_n) + (x_n - y)| \leq |x_n - x| + |x_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $|x - y| < \varepsilon$ for every $\varepsilon > 0$, we must have $|x - y| = 0$, which proves $x = y$. $\square$

2.2 Properties and Algebra of Limits

Definition: Bounded Sequences

A sequence (x_n) is **bounded** if there exists $C > 0$ such that $|x_n| \leq C$ for all $n \in \mathbb{N}$. Otherwise, (x_n) is **unbounded**.

Theorem: Convergent Sequences are Bounded (Theorem 6)

Every convergent sequence of real numbers is bounded.

Proof. Let $x_n \rightarrow x$. Choose $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that $|x_n - x| < 1$ for all $n \geq N$. By the triangle inequality:

$$|x_n| = |(x_n - x) + x| \leq |x_n - x| + |x| < 1 + |x|, \quad \forall n \geq N.$$

Now define $C = \max\{|x_1|, |x_2|, \dots, |x_{N-1}|, 1 + |x|\}$. Then $|x_n| \leq C$ for all $n \in \mathbb{N}$. Hence (x_n) is bounded. $\square$

Remark: The converse is false. The sequence $x_n = (-1)^n$ is bounded ($|x_n| \leq 1$), but diverges due to oscillation.

Theorem: Terms Bounded Away from Zero (Theorem 7)

Let $x_n \rightarrow x$ with $x \neq 0$. Then there exists $N \in \mathbb{N}$ such that $|x_n| > \frac{|x|}{2}$ for all $n \geq N$.

Proof. Choose $\varepsilon = \frac{|x|}{2} > 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that $|x_n - x| < \frac{|x|}{2}$ for all $n \geq N$. By the reverse triangle inequality:

$$|x_n| = |x - (x - x_n)| \geq |x| - |x_n - x| > |x| - \frac{|x|}{2} = \frac{|x|}{2}, \quad \forall n \geq N.$$

$\square$

Theorem: Preservation of Non-negativity (Theorem 8)

Let $a_n \rightarrow a$. If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

Proof. Suppose, on the contrary, that $a < 0$. Set $\delta = -a > 0$. Since $a_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that $a_n \in (a - \delta, a + \delta)$ for all $n \geq N$. Since $a + \delta = a + (-a) = 0$, the interval is $(2a, 0)$, meaning $a_n < 0$ for all $n \geq N$, which contradicts $a_n \geq 0$. Hence $a \geq 0$. $\square$

Theorem: Algebra of Convergent Sequences (Theorem 9)

Let $x_n \rightarrow x$, $y_n \rightarrow y$, and $\alpha \in \mathbb{R}$. Then:

1. $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$.
2. $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x$.
3. $\lim_{n \rightarrow \infty} (x_n y_n) = xy$.
4. If $x_n > 0$ for all n and $x > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}$.

Proof. 1. Given $\varepsilon > 0$, choose N_1, N_2 such that $|x_n - x| < \varepsilon/2$ for $n \geq N_1$ and $|y_n - y| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$: $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \varepsilon$.

2. If $\alpha = 0$, trivial. If $\alpha \neq 0$, choose N such that $|x_n - x| < \frac{\varepsilon}{|\alpha|}$. Then $|\alpha x_n - \alpha x| = |\alpha| |x_n - x| < \varepsilon$.

3. Observe that $|x_n y_n - xy| = |x_n(y_n - y) + y(x_n - x)| \leq |x_n| |y_n - y| + |y| |x_n - x|$. Since (x_n) converges, $\exists N_1$ with $|x_n| < |x| + 1$ for $n \geq N_1$. Choose N_2 with $|y_n - y| < \frac{\varepsilon}{2(|x|+1)}$ and N_3 with $|x_n - x| < \frac{\varepsilon}{2(1+|y|)}$. For $n \geq \max(N_1, N_2, N_3)$, $|x_n y_n - xy| < (|x| + 1) \frac{\varepsilon}{2(|x|+1)} + |y| \frac{\varepsilon}{2(1+|y|)} < \varepsilon$.

4. By Theorem 7, $\exists N_1$ such that $x_n > x/2$. Choose N_2 such that $|x_n - x| < \frac{x^2 \varepsilon}{2}$. For $n \geq \max(N_1, N_2)$:

$$\left| \frac{1}{x_n} - \frac{1}{x} \right| = \frac{|x - x_n|}{xx_n} < \frac{|x_n - x|}{x(x/2)} = \frac{2}{x^2} |x_n - x| < \varepsilon.$$

□

2.3 Cauchy Sequences

Definition: Cauchy Sequence

A sequence of real numbers (x_n) is called a **Cauchy sequence** if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N, |x_n - x_m| < \varepsilon.$$

Theorem: Cauchy Criterion for Sequences (Theorem 10)

A sequence of real numbers is convergent if and only if it is a Cauchy sequence.

Proof. ($\implies$) Let $x_n \rightarrow x$. Given $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon/2$ for all $n \geq N$. Then for $m, n \geq N$:

$$|x_n - x_m| \leq |x_n - x| + |x - x_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

($\impliedby$) Let (x_n) be Cauchy. Define the set:

$$E = \{x \in \mathbb{R} : \exists N \in \mathbb{N} \text{ such that } x < x_n, \forall n \geq N\}.$$

Step 1: For $\delta > 0$, $\exists N_0$ such that $|x_m - x_n| < \delta$ for $m, n \geq N_0$. Taking $m = N_0$, $x_n \in (x_{N_0} - \delta, x_{N_0} + \delta)$ for all $n \geq N_0$. Thus $x_{N_0} - \delta \in E$ (so $E \neq \emptyset$), and $x_{N_0} + \delta$ is an upper bound of E .

Step 2: By completeness of $\mathbb{R}$, $\ell = \sup E$ exists.

Step 3: We prove $x_n \rightarrow \ell$. Let $\varepsilon > 0$. Since (x_n) is Cauchy, $\exists N_0$ such that $|x_n - x_{N_0}| < \varepsilon/2$ for all $n \geq N_0$. Then $x_{N_0} - \varepsilon/2 \in E \implies x_{N_0} - \varepsilon/2 \leq \ell$. Also, $x_{N_0} + \varepsilon/2$ is an upper bound of $E \implies \ell \leq x_{N_0} + \varepsilon/2$. Thus $|x_{N_0} - \ell| \leq \varepsilon/2$.

Then for all $n \geq N_0$:

$$|x_n - \ell| \leq |x_n - x_{N_0}| + |x_{N_0} - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} x_n = \ell$.

□

2.4 Monotone Sequences and the Sandwich Theorem

Definition: Monotonicity

A sequence (x_n) is **increasing** if $x_n \leq x_{n+1}$ for all n ; **strictly increasing** if $x_n < x_{n+1}$; **decreasing** if $x_n \geq x_{n+1}$; **strictly decreasing** if $x_n > x_{n+1}$. A sequence is **monotone** if it is either increasing or decreasing.

Theorem: Monotone Convergence Theorem (Theorem 11)

An increasing sequence of real numbers is convergent if and only if it is bounded above. In this case:

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Similarly, a decreasing sequence is convergent if and only if it is bounded below, and converges to its infimum.

Proof. Let (x_n) be increasing and bounded above. Let $E = \{x_n : n \in \mathbb{N}\}$. By the completeness property of $\mathbb{R}$, $\ell = \sup E$ exists in $\mathbb{R}$.

Let $\varepsilon > 0$. Since $\ell - \varepsilon < \ell$, $\ell - \varepsilon$ is not an upper bound of E . Thus there exists $N \in \mathbb{N}$ such that $x_N > \ell - \varepsilon$.

Since (x_n) is increasing, for every $n \geq N$:

$$\ell - \varepsilon < x_N \leq x_n \leq \ell < \ell + \varepsilon \implies |x_n - \ell| < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} x_n = \ell = \sup E$. The converse follows because all convergent sequences are bounded. $\square$

Theorem: Sandwich (Squeeze) Theorem (Theorem 12)

Let $(x_n), (y_n), (z_n)$ be sequences of real numbers such that $x_n \leq z_n \leq y_n$ for all $n \in \mathbb{N}$. If $\lim_{n \rightarrow \infty} x_n = \alpha = \lim_{n \rightarrow \infty} y_n$, then:

$$\lim_{n \rightarrow \infty} z_n = \alpha.$$

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow \alpha$, $\exists N_1$ such that $\alpha - \varepsilon < x_n < \alpha + \varepsilon$ for all $n \geq N_1$. Since $y_n \rightarrow \alpha$, $\exists N_2$ such that $\alpha - \varepsilon < y_n < \alpha + \varepsilon$ for all $n \geq N_2$.

Let $N = \max\{N_1, N_2\}$. For all $n \geq N$:

$$\alpha - \varepsilon < x_n \leq z_n \leq y_n < \alpha + \varepsilon \implies |z_n - \alpha| < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} z_n = \alpha$. $\square$

Example: Core Illustrative Limits via MCT and Sandwich Theorem

- Geometric sequence:** For $0 \leq r < 1$, $\lim_{n \rightarrow \infty} r^n = 0$. Since $r^{n+1} \leq r^n$ and $r^n \geq 0$, (r^n) converges by MCT to ℓ . Then $\ell = \lim r^{n+1} = r\ell \implies (1-r)\ell = 0 \implies \ell = 0$.
- Oscillating vanishing sequence:** $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$. Since $-1 \leq \sin n \leq 1$, $-1/n \leq \frac{\sin n}{n} \leq 1/n$. By Sandwich Theorem, limit is 0.
- Sum of reciprocal square roots:** $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2+k}} = 1$. Bounded between $\frac{n}{\sqrt{n^2+n}} \rightarrow 1$ and $\frac{n}{\sqrt{n^2+1}} \rightarrow 1$.
- Limit definition of e :** The sequence $a_n = (1 + 1/n)^n$ is strictly increasing (by AM-GM on n terms of $\frac{n+1}{n}$ and one 1) and bounded above by 3 (via Binomial Expansion). By MCT, it converges to $e \approx 2.71828$.

2.5 Sequences Diverging to $\pm\infty$, Ratio and Root Tests

Definition: Divergence to $\pm\infty$

- $x_n \rightarrow +\infty$ if $\forall R \in \mathbb{R}, \exists N \in \mathbb{N}$ such that $\forall n \geq N, x_n > R$.
- $x_n \rightarrow -\infty$ if $\forall R \in \mathbb{R}, \exists N \in \mathbb{N}$ such that $\forall n \geq N, x_n < R$.

Theorem: Ratio and Root Tests for Sequences (Theorems 13 & 14)

Let (u_n) be a sequence of positive real numbers.

1. **Ratio Test:** If $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell$, then:

$$\text{If } 0 \leq \ell < 1 \implies \lim u_n = 0; \quad \text{If } \ell > 1 \implies \lim u_n = +\infty.$$

2. **Root Test:** If $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell$, then:

$$\text{If } 0 \leq \ell < 1 \implies \lim u_n = 0; \quad \text{If } \ell > 1 \implies \lim u_n = +\infty.$$

3. If $\ell = 1$, both tests are **inconclusive** (e.g., $u_n = n \rightarrow \infty$ vs. $u_n = 1/n \rightarrow 0$, both have ratio/root limits of 1).

2.6 Subsequences and the Bolzano–Weierstrass Theorem

Definition: Subsequence

Let (r_n) be a strictly increasing sequence of natural numbers ($r_1 < r_2 < r_3 < \dots$, so $r_n \geq n$). If (x_n) is a sequence, (x_{r_n}) is called a **subsequence** of (x_n) .

Theorem: Subsequence Limit Theorem (Theorem 15)

If a sequence (x_n) converges to x , then every subsequence (x_{r_n}) also converges to x .

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow x$, $\exists N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$. Since $r_n \geq n$, for all $n \geq N$ we have $r_n \geq N$, which implies $|x_{r_n} - x| < \varepsilon$. Thus $x_{r_n} \rightarrow x$. $\square$

Theorem: Existence of Monotone Subsequences & Bolzano–Weierstrass (Theorems 16 & 17)

1. **Theorem 16:** Every sequence of real numbers has a monotone subsequence.
2. **Theorem 17 (Bolzano–Weierstrass):** Every bounded sequence of real numbers has a convergent subsequence.

3 Infinite Series

3.1 Convergence, Cauchy Criterion, and the Divergence Test

An infinite series is a formal sum $\sum_{n=1}^{\infty} a_n$. Its behavior is determined by its sequence of partial sums $s_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$.

- If $\lim_{n \rightarrow \infty} s_n = s \in \mathbb{R}$, the series **converges** and we write $\sum_{n=1}^{\infty} a_n = s$.
- If (s_n) diverges, the series **diverges**.

Theorem: Cauchy Criterion for Series (Theorem 18)

A series $\sum_{n=1}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that:

$$\forall n > m \geq N, \quad |s_n - s_m| = |a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon.$$

Theorem: Divergence Test (n -th Term Test - Theorem 19)

If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Proof. Since $a_n = s_n - s_{n-1}$ for $n \geq 2$, and $\lim s_n = s$, by the algebra of limits:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = s - s = 0.$$

□

Warning: $a_n \rightarrow 0$ does **not** imply convergence! The Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n}$ has $a_n = 1/n \rightarrow 0$, yet diverges to $+\infty$ ($s_{2^m} \geq 1 + m/2 \rightarrow \infty$).

3.2 Absolute Convergence and Convergence Tests

Definition: Absolute vs. Conditional Convergence

- A series $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges.
- A series $\sum a_n$ is **conditionally convergent** if $\sum a_n$ converges but $\sum |a_n|$ diverges (e.g., $\sum \frac{(-1)^{n-1}}{n}$).

Theorem: Absolute Convergence Implies Convergence (Theorem 22)

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof. Let $s_n = \sum_{k=1}^n a_k$ and $\sigma_n = \sum_{k=1}^n |a_k|$. Since (σ_n) converges, it is a Cauchy sequence. For $n > m \geq N$:

$$|s_n - s_m| = \left| \sum_{k=m+1}^n a_k \right| \leq \sum_{k=m+1}^n |a_k| = \sigma_n - \sigma_m < \varepsilon.$$

Thus (s_n) is Cauchy, hence convergent by the Cauchy Criterion.

□

Theorem: Direct Comparison Test (Theorem 23)

Let $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}$.

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Theorem: d'Alembert's Ratio Test and Cauchy's Root Test (Theorems 24 & 25)

Let $a_n > 0$.

1. **Ratio Test:** If $\lim \frac{a_{n+1}}{a_n} = r$: $r < 1 \implies \sum a_n$ converges; $r > 1 \implies \sum a_n$ diverges; $r = 1 \implies$ inconclusive.
2. **Root Test:** If $\lim \sqrt[n]{a_n} = r$: $r < 1 \implies \sum a_n$ converges; $r > 1 \implies \sum a_n$ diverges; $r = 1 \implies$ inconclusive.

Theorem: Abel–Pringsheim Criterion (Theorem 26)

Let (a_n) be a decreasing sequence of non-negative real numbers. If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} na_n = 0$.

Proof. Since $\sum a_n$ converges, for $\varepsilon > 0$, $\exists N$ such that $n > m \geq N \implies \sum_{k=m+1}^n a_k < \varepsilon/2$. For $k \geq N$, since a_n is decreasing:

$$ka_{2k} \leq a_{k+1} + a_{k+2} + \cdots + a_{2k} = s_{2k} - s_k < \frac{\varepsilon}{2} \implies 2ka_{2k} < \varepsilon.$$

Thus $(2n)a_{2n} \rightarrow 0$. For odd terms, $(2n+1)a_{2n+1} \leq (2n+1)a_{2n} = (2n)a_{2n} + a_{2n}$. Since $a_{2n} \rightarrow 0$ and $(2n)a_{2n} \rightarrow 0$, we have $(2n+1)a_{2n+1} \rightarrow 0$. By Exercise 15, $na_n \rightarrow 0$. $\square$

Theorem: Leibniz Test for Alternating Series (Theorem 27)

Let (a_n) be a decreasing sequence of non-negative real numbers such that $\lim_{n \rightarrow \infty} a_n = 0$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ converges. Moreover, its sum S satisfies $a_1 - a_2 \leq S \leq a_1$.

Proof. Let $s_n = \sum_{k=1}^n (-1)^{k-1} a_k$.

- $s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0 \implies (s_{2n})$ is monotonically increasing.
- $s_{2n} = a_1 - (a_2 - a_3) - \cdots - a_{2n} \leq a_1 \implies (s_{2n})$ is bounded above by a_1 .
- By MCT, $s_{2n} \rightarrow S$.
- $s_{2n+1} = s_{2n} + a_{2n+1} \rightarrow S + 0 = S$.

Since even and odd subsequences converge to S , the full sequence $s_n \rightarrow S$. Finally, $a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1$. $\square$

4 Continuity of Functions

4.1 Sequential and ε - δ Definitions

Definition: Continuity of a Function

Let $J \subseteq \mathbb{R}$, $f : J \rightarrow \mathbb{R}$, and $a \in J$.

- **Sequential Definition:** f is continuous at a if for every sequence (x_n) in J with $x_n \rightarrow a$, we have $f(x_n) \rightarrow f(a)$.
- **ε - δ Definition:** f is continuous at a if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } \forall x \in J \text{ with } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

f is continuous on J if it is continuous at every point $a \in J$.

Theorem: Equivalence of Continuity Definitions (Theorem 29)

The sequential definition of continuity and the ε - δ definition are equivalent.

Proof. (ε - $\delta \implies$ **sequential**): Assume the ε - δ condition holds at a . Let (x_n) be any sequence in J such that $x_n \rightarrow a$. Given $\varepsilon > 0$, by the ε - δ condition there exists $\delta > 0$ such that $x \in J, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$. Since $x_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that $|x_n - a| < \delta$ for all $n \geq N$. Consequently, $|f(x_n) - f(a)| < \varepsilon$ for all $n \geq N$. This proves $f(x_n) \rightarrow f(a)$.

(**sequential** $\implies$ ε - δ): Assume sequential continuity holds at a . Suppose, for contradiction, that the ε - δ condition fails. Then:

$$\exists \varepsilon_0 > 0 \text{ such that } \forall \delta > 0, \exists x \in J \text{ with } |x - a| < \delta \text{ and } |f(x) - f(a)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, set $\delta = 1/n$. Then there exists $x_n \in J$ such that:

$$|x_n - a| < \frac{1}{n} \quad \text{and} \quad |f(x_n) - f(a)| \geq \varepsilon_0.$$

Since $|x_n - a| < 1/n$, we have $x_n \rightarrow a$. By sequential continuity, this implies $f(x_n) \rightarrow f(a)$, which means there exists $N \in \mathbb{N}$ such that $|f(x_n) - f(a)| < \varepsilon_0$ for all $n \geq N$. This directly contradicts $|f(x_n) - f(a)| \geq \varepsilon_0$ for all n . Thus the ε - δ condition must hold. $\square$

4.2 Sign Preservation and the Intermediate Value Theorem

Theorem: Sign Preservation of Continuous Functions (Theorem 30)

Let $J \subseteq \mathbb{R}$ and let $f : J \rightarrow \mathbb{R}$ be continuous at $c \in J$. If $f(c) \neq 0$, then there exists $\delta > 0$ such that:

$$|f(x)| > \frac{|f(c)|}{2}, \quad \forall x \in (c - \delta, c + \delta) \cap J.$$

In particular, if $f(c) > 0$, then $f(x) > \frac{f(c)}{2} > 0$ on that neighborhood.

Proof. Choose $\varepsilon = \frac{|f(c)|}{2} > 0$. By continuity, there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta) \cap J$, $|f(x) - f(c)| < \frac{|f(c)|}{2}$. By the reverse triangle inequality:

$$|f(x)| = |f(c) - (f(c) - f(x))| \geq |f(c)| - |f(x) - f(c)| > |f(c)| - \frac{|f(c)|}{2} = \frac{|f(c)|}{2}.$$

If $f(c) > 0$, then $f(c) - f(x) \leq |f(x) - f(c)| < \frac{f(c)}{2} \implies f(x) > f(c) - \frac{f(c)}{2} = \frac{f(c)}{2} > 0$. $\square$

Theorem: Intermediate Value Theorem (IVT - Theorems 31 & 32)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous.

1. **Root Existence:** If $f(a) < 0 < f(b)$ (or $f(a) > 0 > f(b)$), then there exists $c \in (a, b)$ such that $f(c) = 0$.
2. **Standard Form:** If $\lambda \in \mathbb{R}$ is strictly between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that $f(c) = \lambda$.

Proof. Proof of Root Existence: Assume $f(a) < 0 < f(b)$. Define the set:

$$E = \{x \in [a, b] : f(y) \leq 0 \text{ for all } y \in [a, x]\}.$$

E is nonempty since $a \in E$, and E is bounded above by b . By the completeness property of $\mathbb{R}$, $c = \sup E$ exists in $[a, b]$.

- Suppose $f(c) < 0$. Then $c < b$ (since $f(b) > 0$). By Theorem 30, $\exists \delta > 0$ such that $f(x) < 0$ on $(c - \delta, c + \delta) \cap [a, b]$. This allows us to extend the set E to $c + \delta/2$, contradicting that c is an upper bound of E .
- Suppose $f(c) > 0$. Then $c > a$ (since $f(a) < 0$). By Theorem 30, $\exists \delta > 0$ such that $f(x) > 0$ on $(c - \delta, c + \delta) \cap [a, b]$. Then no points in $(c - \delta, c]$ can belong to E , meaning $c - \delta$ is an upper bound of E , contradicting $c = \sup E$.

Therefore, we must have $f(c) = 0$. Since $f(a) < 0$ and $f(b) > 0$, $c \in (a, b)$.

Proof of Standard Form: Define $h(x) = f(x) - \lambda$. Then h is continuous, and $h(a)$ and $h(b)$ have opposite signs. Applying Part (1) to h gives $h(c) = 0 \implies f(c) = \lambda$. $\square$

5 Complete Step-by-Step Solutions to All Practice Problems

Exercise / Problem Solution: Exercise 1: Definition of Lower Bound

Problem Statement: Define the notion of a lower bound for a nonempty subset of $\mathbb{R}$.

Solution:

Let $A \subset \mathbb{R}$ be a nonempty subset. A real number $\beta \in \mathbb{R}$ is called a **lower bound** of A if for every $x \in A$, we have $\beta \leq x$.

In quantifier notation: $\forall x \in A (\beta \leq x)$.

Geometric Interpretation: On the real number line, every point in A lies to the right of or coincides with the point β .

Exercise / Problem Solution: Exercise 2: Bounds on Natural Numbers $\mathbb{N}$

Problem Statement: Show that $\mathbb{N}$ has a lower bound in $\mathbb{R}$. Does $\mathbb{N}$ have an upper bound in $\mathbb{R}$?

Solution:

1. **Lower bound for $\mathbb{N}$:** For every $n \in \mathbb{N} = \{1, 2, 3, \dots\}$, we have $1 \leq n$. Thus $\beta = 1$ is a lower bound for $\mathbb{N}$. Any real number $\beta \leq 1$ (e.g., $0, -1, -\pi$) is also a lower bound. In fact, $\inf \mathbb{N} = \min \mathbb{N} = 1$.
2. **Upper bound for $\mathbb{N}$:** $\mathbb{N}$ has **no upper bound** in $\mathbb{R}$. As proven in Theorem 2, if $\mathbb{N}$ were bounded above, then by the Completeness Property, $\alpha = \sup \mathbb{N} \in \mathbb{R}$ would exist. Then $\alpha - 1 < \alpha$ would imply the existence of $n \in \mathbb{N}$ such that $n > \alpha - 1 \implies n + 1 > \alpha$, contradicting that α is an upper bound.

Exercise / Problem Solution: Exercise 3: Sets Bounded and Not Bounded Below

Problem Statement: Using the definition of bounded above, define what it means for a subset of $\mathbb{R}$ to be bounded below. Visualize on the number line. When do you say that a subset $A \subset \mathbb{R}$ is not bounded below in $\mathbb{R}$?

Solution:

1. **Bounded Below:** A nonempty subset $A \subset \mathbb{R}$ is *bounded below* in $\mathbb{R}$ if there exists $\beta \in \mathbb{R}$ such that $\beta \leq x$ for all $x \in A$, written:

$$\exists \beta \in \mathbb{R} (\forall x \in A (\beta \leq x)).$$

2. **Visualization:** There is a vertical barrier at β such that no elements of A spill over to the left of β .
3. **Not Bounded Below:** A is *not bounded below* if no real number β can serve as a lower bound:

$$\forall \beta \in \mathbb{R}, \exists x \in A \text{ such that } x < \beta.$$

Exercise / Problem Solution: Exercise 4: Uniqueness of Maximum

Problem Statement: Suppose that $M, M' \in A$ are both maximums of a nonempty subset $A \subset \mathbb{R}$. Prove that $M = M'$.

Solution:

Proof. By definition of maximum:

- Since M is a maximum of A , M is an upper bound of A . Since $M' \in A$, we must have $M' \leq M$.
- Since M' is a maximum of A , M' is an upper bound of A . Since $M \in A$, we must have $M \leq M'$.

Combining $M' \leq M$ and $M \leq M'$ yields $M = M'$. □

Exercise / Problem Solution: Exercise 5: Minimum of a Set and Its Uniqueness

Problem Statement: Define the minimum of a nonempty subset $A \subset \mathbb{R}$. Prove that if A has a minimum, then it is unique.

Solution:

1. **Definition:** An element $m \in \mathbb{R}$ is called the **minimum** of A (denoted $m = \min A$) if:

- (i) $m \in A$;
- (ii) m is a lower bound of A , i.e., $m \leq x$ for all $x \in A$.

2. **Proof of Uniqueness:** Suppose $m_1, m_2 \in A$ are both minimums of A .

- Since m_1 is a lower bound and $m_2 \in A$, we have $m_1 \leq m_2$.
- Since m_2 is a lower bound and $m_1 \in A$, we have $m_2 \leq m_1$.

Hence $m_1 = m_2$.

Exercise / Problem Solution: Exercise 6: Gap in $\mathbb{Q}$ (Non-Completeness)

Problem Statement: Construct a nonempty subset of $\mathbb{Q}$ that is bounded above but whose least upper bound does not belong to $\mathbb{Q}$.

Solution:

Consider the set:

$$A = \{x \in \mathbb{Q} : x > 0 \text{ and } x^2 < 2\}.$$

- A is nonempty because $1 \in \mathbb{Q}$ and $1^2 = 1 < 2$, so $1 \in A$.
- A is bounded above in $\mathbb{Q}$ by $2 \in \mathbb{Q}$, since if $x \in \mathbb{Q}$ and $x > 2$, then $x^2 > 4 > 2$, so $x \notin A$.
- In the real number system $\mathbb{R}$, $\sup A = \sqrt{2}$.
- Since $\sqrt{2} \notin \mathbb{Q}$ (it is irrational), the set A has no least upper bound in the field $\mathbb{Q}$.

Exercise / Problem Solution: Exercise 7: Bounds on Integers $\mathbb{Z}$

Problem Statement: Prove that the set of integers $\mathbb{Z}$ is neither bounded above nor bounded below in $\mathbb{R}$.

Solution:

Proof. 1. **$\mathbb{Z}$ is not bounded above:** We know $\mathbb{N} \subset \mathbb{Z}$. If $\mathbb{Z}$ were bounded above by some $M \in \mathbb{R}$, then for all $n \in \mathbb{N} \subset \mathbb{Z}$, we would have $n \leq M$. This would make M an upper bound for $\mathbb{N}$, directly contradicting Theorem 2. Thus $\mathbb{Z}$ is unbounded above.

2. **$\mathbb{Z}$ is not bounded below:** Suppose for contradiction that $\mathbb{Z}$ is bounded below by some $m \in \mathbb{R}$. Then $m \leq k$ for all $k \in \mathbb{Z}$. For every $n \in \mathbb{N}$, $-n \in \mathbb{Z}$, which implies $m \leq -n \iff n \leq -m$. This means $-m$ is an upper bound for $\mathbb{N}$, again contradicting Theorem 2. Thus $\mathbb{Z}$ is unbounded below. $\square$

Exercise / Problem Solution: Exercise 8: Limit of $x_n = \frac{1}{an+b}$

Problem Statement: Using Example 10, show that if $a, b \in \mathbb{R}$ with $a \neq 0$, then the sequence $x_n = \frac{1}{an+b}$ converges. Determine its limit.

Solution:

We claim that $\lim_{n \rightarrow \infty} \frac{1}{an+b} = 0$.

Proof. Let $\varepsilon > 0$ be given. We want to show $|x_n - 0| = \frac{1}{|an+b|} < \varepsilon$, which is equivalent to:

$$|an + b| > \frac{1}{\varepsilon}.$$

By the reverse triangle inequality: $|an + b| \geq |an| - |b| = |a|n - |b|$.

To achieve $|an + b| > 1/\varepsilon$, it suffices to make $|a|n - |b| > 1/\varepsilon$, which is equivalent to:

$$|a|n > |b| + \frac{1}{\varepsilon} \iff n > \frac{|b| + 1/\varepsilon}{|a|}.$$

Since $a \neq 0$, the quantity $\frac{|b| + 1/\varepsilon}{|a|}$ is a well-defined real number. By the Archimedean Property, there exists $N \in \mathbb{N}$ such that $N > \frac{|b| + 1/\varepsilon}{|a|}$.

Then for all $n \geq N$:

$$|an + b| \geq |a|n - |b| \geq |a|N - |b| > \frac{1}{\varepsilon} \implies \left| \frac{1}{an+b} - 0 \right| < \varepsilon.$$

Hence $\lim_{n \rightarrow \infty} \frac{1}{an+b} = 0$. $\square$

Exercise / Problem Solution: Exercise 9: Inequality of Limits

Problem Statement: Let (u_n) and (v_n) be two convergent sequences such that $u_n \leq v_n$ for all $n \in \mathbb{N}$. Show that $\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n$.

Solution:

Proof. Let $\lim_{n \rightarrow \infty} u_n = u$ and $\lim_{n \rightarrow \infty} v_n = v$. Define the sequence $w_n = v_n - u_n$. Since $u_n \leq v_n$, we have $w_n = v_n - u_n \geq 0$ for all $n \in \mathbb{N}$. By the Algebra of Limits (Theorem 9), (w_n) converges, and:

$$\lim_{n \rightarrow \infty} w_n = \lim_{n \rightarrow \infty} v_n - \lim_{n \rightarrow \infty} u_n = v - u.$$

Since $w_n \geq 0$ for all n , Theorem 8 (preservation of non-negativity) implies that its limit is non-negative:

$$v - u \geq 0 \implies u \leq v.$$

Therefore, $\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n$. □

Exercise / Problem Solution: Exercise 10: Counterexamples for $x = 0$ in Reciprocal Limit

Problem Statement: In Theorem 9(4), the assumption $x > 0$ is essential. What can go wrong if $x = 0$? Give an example to show that the conclusion need not hold.

Solution:

If $\lim_{n \rightarrow \infty} x_n = 0$, the sequence of reciprocals $\left(\frac{1}{x_n}\right)$ cannot converge to any real number in $\mathbb{R}$.

1. **Example 1 (Divergence to $+\infty$):** Let $x_n = \frac{1}{n} > 0$. Then $\lim x_n = 0$. However, $\frac{1}{x_n} = n$, which is unbounded and diverges to $+\infty$.
2. **Example 2 (Oscillating unbounded divergence):** Let $x_n = \frac{(-1)^n}{n}$. Then $\lim x_n = 0$. However, $\frac{1}{x_n} = (-1)^n n = (-1, 2, -3, 4, -5, \dots)$, which oscillates wildly and does not converge to any real number or even to $\pm\infty$.

Thus the operation $1/x$ is undefined when $x = 0$.

Exercise / Problem Solution: Exercise 11: MCT for Decreasing Sequences

Problem Statement: Show that a decreasing sequence of real numbers is convergent if and only if it is bounded below.

Solution:

Proof. ($\implies$) If (x_n) is convergent, it is bounded (Theorem 6), hence bounded below. ($\impliedby$) Let (x_n) be a decreasing sequence ($x_{n+1} \leq x_n$ for all n) that is bounded below. Let $E = \{x_n : n \in \mathbb{N}\}$.

- Since $E \neq \emptyset$ and E is bounded below, by the completeness of $\mathbb{R}$ (infimum property), $\ell = \inf E$ exists in $\mathbb{R}$.
- Let $\varepsilon > 0$ be given. Since $\ell + \varepsilon > \ell$, $\ell + \varepsilon$ cannot be a lower bound for E . Thus there exists $N \in \mathbb{N}$ such that $x_N < \ell + \varepsilon$.
- Since (x_n) is decreasing, for every $n \geq N$: $x_n \leq x_N < \ell + \varepsilon$.
- Furthermore, since $\ell = \inf E$ is a lower bound, $\ell - \varepsilon < \ell \leq x_n$.
- Combining these inequalities: $\ell - \varepsilon < x_n < \ell + \varepsilon \iff |x_n - \ell| < \varepsilon$ for all $n \geq N$.

Therefore, $\lim_{n \rightarrow \infty} x_n = \ell = \inf E$. □

Exercise / Problem Solution: Exercise 12: Bounded Times Null Sequence and Power Limit**Problem Statement:**

1. Let (a_n) be a bounded sequence of real numbers and let (x_n) be a sequence such that $x_n \rightarrow 0$. Show that $\lim_{n \rightarrow \infty} a_n x_n = 0$.
2. Deduce that $\lim_{n \rightarrow \infty} r^n = 0$ for every $|r| < 1$.

Solution:

1. Since (a_n) is bounded, there exists $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$. Since $x_n \rightarrow 0$, there exists $N \in \mathbb{N}$ such that $|x_n - 0| = |x_n| < \frac{\varepsilon}{M}$ for all $n \geq N$. Then for all $n \geq N$:

$$|a_n x_n - 0| = |a_n| |x_n| \leq M |x_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Therefore, $\lim_{n \rightarrow \infty} a_n x_n = 0$.

2. If $r = 0$, $r^n = 0 \rightarrow 0$. If $0 < |r| < 1$, write:

$$r^n = |r|^n \left(\frac{r}{|r|} \right)^n.$$

Let $x_n = |r|^n$ and $a_n = \left(\frac{r}{|r|} \right)^n$. Since $0 < |r| < 1$, Example 21 guarantees that $x_n = |r|^n \rightarrow 0$. Moreover, $|a_n| = 1^n = 1$, so (a_n) is bounded. By part (1), $\lim_{n \rightarrow \infty} r^n = \lim_{n \rightarrow \infty} a_n x_n = 0$.

Exercise / Problem Solution: Exercise 13: Core Fundamental Limit Theorems**Problem Statement:** Solve the following five fundamental limit problems:

1. Show that $\lim_{n \rightarrow \infty} n^{1/n} = 1$.
2. Let $a > 1$. Show that $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$.
3. Let $a \in \mathbb{R}$. Show that $\lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0$.
4. Let (x_n) be a sequence converging to 0. Define $s_n := \frac{x_1 + x_2 + \dots + x_n}{n}$. Show that $s_n \rightarrow 0$.
5. Let $x_n \rightarrow x$ and $y_n \rightarrow y$. Show that $\frac{x_1 y_n + x_2 y_{n-1} + \dots + x_n y_1}{n} \rightarrow xy$.

Solution:

1. For $n \geq 2$, $n^{1/n} > 1$. Let $n^{1/n} = 1 + k_n$ where $k_n > 0$. By the Binomial Theorem:

$$n = (1 + k_n)^n = 1 + nk_n + \frac{n(n-1)}{2} k_n^2 + \dots + k_n^n > \frac{n(n-1)}{2} k_n^2.$$

Dividing by $\frac{n(n-1)}{2}$ gives $k_n^2 < \frac{2}{n-1} \implies 0 < k_n < \sqrt{\frac{2}{n-1}}$. Since $\sqrt{\frac{2}{n-1}} \rightarrow 0$, by Sandwich Theorem $k_n \rightarrow 0$, hence $n^{1/n} = 1 + k_n \rightarrow 1$.

2. Since $a > 1$, write $a = 1 + h$ with $h > 0$. For $n \geq 2$, by Binomial Theorem:

$$a^n = (1 + h)^n > \frac{n(n-1)}{2} h^2.$$

Hence: $0 < \frac{n}{a^n} < \frac{n}{\frac{n(n-1)}{2}h^2} = \frac{2}{(n-1)h^2}$. As $n \rightarrow \infty$, $\frac{2}{(n-1)h^2} \rightarrow 0$, so $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$.

3. If $a = 0$, trivial. For $a \neq 0$, choose $K \in \mathbb{N}$ such that $K > 2|a|$. For all $n > K$:

$$\left| \frac{a^n}{n!} \right| = \frac{|a|^K}{K!} \cdot \frac{|a|}{K+1} \cdot \frac{|a|}{K+2} \cdots \frac{|a|}{n} \leq \frac{|a|^K}{K!} \left(\frac{1}{2} \right)^{n-K} = C \left(\frac{1}{2} \right)^n \rightarrow 0.$$

4. Let $\varepsilon > 0$. Since $x_n \rightarrow 0$, $\exists K \in \mathbb{N}$ such that $|x_n| < \frac{\varepsilon}{2}$ for all $n > K$. For $n > K$:

$$|s_n| \leq \frac{|x_1 + \cdots + x_K|}{n} + \frac{|x_{K+1}| + \cdots + |x_n|}{n} < \frac{\sum_{j=1}^K |x_j|}{n} + \frac{(n-K)(\varepsilon/2)}{n} < \frac{\sum_{j=1}^K |x_j|}{n} + \frac{\varepsilon}{2}.$$

Since K is fixed, the numerator $\sum_{j=1}^K |x_j|$ is constant. Choose $N > K$ such that for $n \geq N$, $\frac{\sum_{j=1}^K |x_j|}{n} < \frac{\varepsilon}{2}$. Then $|s_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$, so $s_n \rightarrow 0$.

5. Write $x_n = x + u_n$ and $y_n = y + v_n$, where $u_n \rightarrow 0$ and $v_n \rightarrow 0$. The Cauchy mean expands to:

$$\frac{1}{n} \sum_{k=1}^n x_k y_{n-k+1} = xy + x \left(\frac{1}{n} \sum_{k=1}^n v_k \right) + y \left(\frac{1}{n} \sum_{k=1}^n u_k \right) + \frac{1}{n} \sum_{k=1}^n u_k v_{n-k+1}.$$

By Part (4), the Cesàro averages of u_k and v_k tend to 0. Since (u_k) is bounded ($|u_k| \leq M$), $\left| \frac{1}{n} \sum u_k v_{n-k+1} \right| \leq M \left(\frac{1}{n} \sum |v_j| \right) \rightarrow 0$. Thus the total limit is xy .

Exercise / Problem Solution: Exercise 14: Monotonicity and Euler's Constant e

Problem Statement: Let $x_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!}$.

1. Show that (x_n) is monotonically increasing.
2. Show that (x_n) is bounded above.
3. Deduce that (x_n) is convergent.
4. Show that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} (1 + 1/n)^n = e$.

Solution:

1. $x_{n+1} - x_n = \frac{1}{(n+1)!} > 0$ for all $n \in \mathbb{N}$, so (x_n) is strictly increasing.
2. For $k \geq 1$, $k! = 1 \cdot 2 \cdot 3 \cdots k \geq 1 \cdot 2 \cdot 2 \cdots 2 = 2^{k-1}$. Thus:

$$x_n = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!} \leq 1 + 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} < 1 + \frac{1}{1 - 1/2} = 3.$$

Thus (x_n) is bounded above by 3.

3. Since (x_n) is increasing and bounded above, by the Monotone Convergence Theorem (Theorem 11), (x_n) converges to a real limit $S \leq 3$.
4. Let $a_n = (1 + 1/n)^n$. By the Binomial Theorem:

$$a_n = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n} \right) \left(1 - \frac{2}{n} \right) \cdots \left(1 - \frac{k-1}{n} \right).$$

Clearly, each term in the sum is $\leq \frac{1}{k!}$, so $a_n \leq x_n$, which gives $e = \lim a_n \leq S$.
Conversely, fix any integer $m \leq n$:

$$a_n \geq \sum_{k=0}^m \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Taking $n \rightarrow \infty$ with m fixed yields $e \geq \sum_{k=0}^m \frac{1}{k!} = x_m$.

Now letting $m \rightarrow \infty$ gives $e \geq S$. Since $e \leq S$ and $e \geq S$, we conclude $S = e$.

Exercise / Problem Solution: Exercise 15: Convergence via Even and Odd Subsequences

Problem Statement: Let (x_n) be a sequence. Suppose both (x_{2n}) and (x_{2n-1}) converge to the same limit x . Show that (x_n) converges to x .

Solution:

Proof. Let $\varepsilon > 0$.

- Since $x_{2k} \rightarrow x$, $\exists K_1 \in \mathbb{N}$ such that $|x_{2k} - x| < \varepsilon$ for all $k \geq K_1$.
- Since $x_{2k-1} \rightarrow x$, $\exists K_2 \in \mathbb{N}$ such that $|x_{2k-1} - x| < \varepsilon$ for all $k \geq K_2$.

Set $N = \max\{2K_1, 2K_2 - 1\}$. Let $n \geq N$.

- If n is even, $n = 2k$. Then $2k \geq N \geq 2K_1 \implies k \geq K_1$, so $|x_n - x| = |x_{2k} - x| < \varepsilon$.
- If n is odd, $n = 2k - 1$. Then $2k - 1 \geq N \geq 2K_2 - 1 \implies k \geq K_2$, so $|x_n - x| = |x_{2k-1} - x| < \varepsilon$.

In both cases, $|x_n - x| < \varepsilon$ for all $n \geq N$. Hence $\lim_{n \rightarrow \infty} x_n = x$. $\square$

Exercise / Problem Solution: Exercise 16: Bolzano–Weierstrass Applications

Problem Statement:

1. Show that the sequence $(\sin n)$ has a convergent subsequence.
2. Prove that every bounded sequence has a monotone convergent subsequence.
3. True or False: A sequence (x_n) is bounded if and only if every subsequence of (x_n) has a convergent subsequence.

Solution:

1. For all $n \in \mathbb{N}$, $-1 \leq \sin n \leq 1$, so $(\sin n)$ is a bounded real sequence. By the Bolzano–Weierstrass Theorem (Theorem 17), $(\sin n)$ must contain a convergent subsequence.
2. By Theorem 16, every sequence has a monotone subsequence (x_{r_n}) . If (x_n) is bounded, then (x_{r_n}) is also bounded. By the Monotone Convergence Theorem (Theorem 11), any bounded monotone sequence converges. Thus (x_{r_n}) is a monotone convergent subsequence.

3. **Statement is TRUE.**

- ($\implies$) If (x_n) is bounded, any subsequence (y_k) is also bounded, and hence by Bolzano–Weierstrass has a convergent subsequence.
- ($\impliedby$) Contrapositive: If (x_n) is unbounded, then for every $k \in \mathbb{N}$, there exists $n_k > n_{k-1}$ such that $|x_{n_k}| > k$. The subsequence (x_{n_k}) satisfies $|x_{n_k}| \rightarrow \infty$, so every subsequence of (x_n) is also unbounded and cannot converge.

Exercise / Problem Solution: Exercise 17: Abel's Test for Series

Problem Statement: Assume $\sum a_n$ is convergent and (b_n) is monotone and bounded ($m \leq b_n \leq M$ for all n). Prove that $\sum a_n b_n$ is convergent.

Solution:

Proof. Let $s_n = \sum_{k=1}^n a_k$ with $s_0 = 0$. Since $\sum a_n$ converges to s , (s_n) is bounded, say $|s_n| \leq K$ for all n . Using Abel's summation by parts:

$$\sum_{k=1}^n a_k b_k = \sum_{k=1}^n (s_k - s_{k-1}) b_k = s_n b_n + \sum_{k=1}^{n-1} s_k (b_k - b_{k+1}).$$

1. Since (b_n) is bounded and monotone, by MCT $\lim_{n \rightarrow \infty} b_n = b$ exists. Since $s_n \rightarrow s$, $\lim s_n b_n = sb$.
2. Since (b_k) is monotone, $(b_k - b_{k+1})$ never changes sign. Thus:

$$\sum_{k=1}^{n-1} |s_k (b_k - b_{k+1})| \leq K \sum_{k=1}^{n-1} |b_k - b_{k+1}| = K |b_1 - b_n| \leq K(|b_1| + \max(|m|, |M|)).$$

Thus the series $\sum s_k (b_k - b_{k+1})$ is absolutely convergent, hence convergent.

Therefore, the sequence of partial sums of $\sum a_n b_n$ converges. $\square$

Exercise / Problem Solution: Exercise 18: ε - δ Continuity of $f(x) = x^2$ on $\mathbb{R}$

Problem Statement: Show that $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is continuous on $\mathbb{R}$ using the ε - δ definition.

Solution:

Proof. Let $a \in \mathbb{R}$ and let $\varepsilon > 0$ be given. We examine:

$$|f(x) - f(a)| = |x^2 - a^2| = |x - a||x + a|.$$

To bound $|x + a|$, we first require $\delta \leq 1$. Then whenever $|x - a| < \delta \leq 1$:

$$|x| = |(x - a) + a| \leq |x - a| + |a| < 1 + |a|.$$

Consequently: $|x + a| \leq |x| + |a| < 1 + 2|a|$.

Now, choose:

$$\delta = \min \left\{ 1, \frac{\varepsilon}{1 + 2|a|} \right\}.$$

Then whenever $|x - a| < \delta$, we have $|x - a| < 1$ (so $|x + a| < 1 + 2|a|$) and $|x - a| < \frac{\varepsilon}{1+2|a|}$. Therefore:

$$|f(x) - f(a)| = |x - a||x + a| < \left(\frac{\varepsilon}{1+2|a|}\right)(1+2|a|) = \varepsilon.$$

Hence $f(x) = x^2$ is continuous at a . Since $a \in \mathbb{R}$ was arbitrary, f is continuous on $\mathbb{R}$. $\square$

Exercise / Problem Solution: Exercise 19: Continuous Function with Integer Range is Constant

Problem Statement: Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function whose range is contained in $\mathbb{Z}$, then f is constant.

Solution:

Proof. Suppose for contradiction that f is not constant. Then there exist two real numbers $x_1 < x_2$ such that $f(x_1) \neq f(x_2)$.

Since the range of f is contained in $\mathbb{Z}$, let $n_1 = f(x_1) \in \mathbb{Z}$ and $n_2 = f(x_2) \in \mathbb{Z}$ with $n_1 \neq n_2$. Without loss of generality, assume $n_1 < n_2$.

Choose any non-integer number λ strictly between n_1 and n_2 , for example:

$$\lambda = n_1 + \frac{1}{2}.$$

Then $\min\{f(x_1), f(x_2)\} = n_1 < \lambda < n_2 = \max\{f(x_1), f(x_2)\}$.

Since f is continuous on the closed interval $[x_1, x_2]$, by the Intermediate Value Theorem (Theorem 32), there must exist some $c \in (x_1, x_2)$ such that:

$$f(c) = \lambda = n_1 + \frac{1}{2}.$$

However, $f(\mathbb{R}) \subseteq \mathbb{Z}$, which requires $f(c) \in \mathbb{Z}$. This means $n_1 + \frac{1}{2} \in \mathbb{Z}$, which is a blatant contradiction since n_1 is an integer and $\frac{1}{2}$ is not.

Therefore, no such pair of points with distinct values can exist, and f must be constant on $\mathbb{R}$. $\square$